MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION, March 2026

Fourth Semester

MG4DSECOM205- ESSENTIALS OF SUPPLY CHAIN MANAGEMENT

(2024 Admission onwards)

Duration: 1 Hour 15 minutes Maximum Marks: 70

Remember(K), Understand (U), Apply(A), Analyses (An), Evaluate(E), Create (C), Skill(S), Interest(I) & Appreciation (Ap).

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

MCQ Questions

Answer any 20 questions

Each question carries 1 mark

1. Which activity is part of the Strategic Design phase of strategic implementation? [U]/[CO3]

a) Training employees on new technologies.
b) Defining clear, measurable objectives (KPIs).
c)Determining the optimal location and number of facilities (Net Design).
d) Thoroughly analysing the current supply Chain.

2. In XYZ Analysis classifies managing items with a fluctuating demand, are classified as: [E]/[CO2]

a) X-Items.
b) Z-Items.
c) V-Items.
d)Y-Items.

3. A company that competes primarily on low cost may favour which type of distribution network to leverage economies of scale in transportation and inventory? [E]/[CO3]

a) A decentralized network.
b) An indirect network only.
c)A centralized network.
d)A hybrid network.

4. Purchasing of material from suppliers in other countries refers to: [E]/[CO4]

a) Domestic sourcing
b) Global sourcing
c)Insourcing
d) Tactical sourcing

5. Which of the following best defines a supply chain? [U]/[CO1]

a) The chain of command in a company
b) A system of advertising channels
c) A network connecting suppliers and consumers through activities and resources
d) A financial chain for product costing

6. The primary goal of a supply chain is to: [U]/[CO1]

a) Increase production cost
b) Create advertisements for products
c) Maximize customer value and achieve sustainable competitive advantage
d) Reduce product quality

7. The purpose of FSN Analysis is to classify inventory based on its consumption rate or turnover, which primarily helps in managing: [E]/ [CO2]

a) Price fluctuations and supplier quality.
b) Obsolescence and storage space.
c)Last-mile delivery costs and lead time.
d)Advertising expenditure and sales forecasts.

8. The Hybrid (Push-Pull) Strategy switches to the Pull system for which stages of the supply chain? [E]/[CO3]

a) The initial stages of producing generic components.
b) The final, customized stages (e.g., assembling the final product).
c) Sourcing and supplier management.
d)Long-term network design

9. Which of the following technology improves real-time visibility in Supply chains? [E]/[CO4]

a) Fax machines
b) Typewriters
c)Internet of Things
d) Manual ledgers

10. The scope of supply chain includes: [U]/[CO1]

a) Only the transportation of goods
b) Only after-sales service
c) Planning, sourcing, production, logistics customer service.
d) Only marketing activities

11. Which of the following activities is NOT typically part of the supply chain? [U]/[CO1]

a) Procurement b) Manufacturing
c) Distribution d) Tax auditing

12. SOS (Seasonal, Off-Seasonal) analysis is a technique used to manage inventory for products with: [E]/[CO2]

a) High value and low volume.
b) Very irregular, low-volume demand.
c) Seasonal demand.
d) Consistent Predictable demand

13. Kaizen in Lean SCM refers to: [E]/[CO4]

a) Outsourcing b) Increased lead time

c) Continuous improvement d) Excessive documentation

14. A company switching to bio degradable packaging is practicing: [E]/[CO4]

a) Lean SCM b) Green SCM
c)Push strategy d) Price discounting

15. Which of the following is NOT a key component of distribution in supply chain management? [E]/[CO3]

a) Warehousing. b) Human Resources Management.
c) Transportation. d) Inventory Management.

16. Which type of supply chain is designed to maximise the cost efficiency in high volume industries? [U]/[CO1]

a) Agile b) Efficient
c)Custom configured d) Flexible

17. The Periodic Review System (P-System) is characterized by: [U]/[CO2]

a) Inventory levels being constantly monitored.
b) Placing a fixed quantity order when stock drops to a reorder point.
c) Checking inventory levels at fixed, predetermined intervals.
d) Placing an order only when demand is highly irregular.

18. Scheduling is the process of assigning: [E]/[CO3]

a) The final sequence of stops for a vehicle.
b) Resources-vehicles, drivers, and time slots-to the routes.
c) The minimum distance between stops.
d) The cargo capacity constraints.

19. Which practice strengthens partnerships by aligning goals? [E]/[CO4]

a) Individual profit maximization b) Shared performance metrics
c)Order batching d) Frequent price promotions

20. Which inventory system is suitable for high-value items because its inventory levels are constantly monitored and an order for a fixed quantity (like EOQ) is placed when stock drops to a reorder point? [E]/[CO3]

a) Periodic Review System (P-System).
b) Continuous Review System (Q-System).
c) Modified Control System.
d) FSN Analysis.

21. Zara's Supply chain strategy is an example of: [U]/[CO1]

a) Agile SC b) Lean SC
c)Cost based c) Push SC

22. Which of the following is the financial performance indicator of Supply Chain? [U]/[CO2]

a) Order accuracy b) Inventory turnover ratio
c)Lead time d) On time delivery

PART B
MCQ Questions
Answer any 25 questions
Each question carries 2 marks

23. For products with a short-expected delivery time (high customer expectation for Response Time), the necessary distribution network design would be a: [E]/[CO3]

a) Centralized network with few warehouses.
b) Decentralized network with many warehouses located close to customers.
c) Hybrid network focusing on low-value items.
d) Direct distribution model only

24. The bullwhip effects results from: [E]/[CO4]

a) Over-coordination
b) Lack of information sharing
c)High product quality
d) Efficient ordering

25. The aspect of co-ordination that ensures suppliers and buyers plan together is : [E]/[CO4]

a) Joint planning
b) Promotion management
c)Order batching
c) Price Discounting

26. Consider the following statement of Assertion(A) and Reasoning(R). Choose the correct option. [U]/[CO1]

Assertion (A): Effective supply chain management enhances customer satisfaction and competitive advantages.

Reasoning (R): It ensures timely availability of quality products and efficient usage of resources.

a) Both A and R are true, and R is the correct explanation of A
b) Both A and R are true, but R is not the correct explanation of A
c) A is true, but R is false
d) A is false, but R is true

27. A company that competes primarily on low cost may favour which type of distribution network to leverage economies of scale in transportation and inventory? [U]/[CO2]

a) A decentralized network.
b) An indirect network only.
c) A centralized network.
d) A hybrid network.

28. In a Direct Distribution network, the manufacturer sells and delivers the product directly to the end customer without using any middlemen, A key advantage of this is: [E]/[CO3]

a) Reduced need for capital investment in infrastructure.
b) Lower overall inventory costs due to larger safety stock across multiple locations.
c) Full control over the customer experience, brand messaging, and pricing.
d) Reduced transportation costs for last-mile delivery.

29. Consider the following statement and choose the most appropriate alternative: [E]/[CO4]

Statement I: Behavioural obstacles include lack of trust and blame shifting among SC partners. Statement II: Behavioural obstacles occur due to misalignment in inventory levels.

a) Both the statements are true
b) Statement I is true, and Statement II is false.

c) Statement I is false, and Statement II is true.
d) Both the statements are false

30. Which of the following inventory analysis method is used for managing spare parts and classify them based on their criticality in production? [U]/[CO2]

a) XYZ Analysis b) VED Analysis
c)ABC Analysis d) FSN Analysis

31. Consider the following statement and choose the most appropriate alternative: [U]/[CO1]
Statement I: The Cash-to-cash cycle time measures the efficiency of cash flow in SC.
Statement II: A longer Cash-to-cash cycle indicates better working capital management.

a) Both the statements are true
b) Statement I is true, and Statement II is false.
c) Statement I is false, and Statement II is true.
d) Both the statements are false

32. Which of the following is a significant benefit of implementing JIT System? [U]/[CO2]

a) High capital tied up in inventory b) Increased need for larger warehouses
c)Reduced inventory cost. d) Increased buffer stock

33. The Pull Strategy is characterized by: [E]/[CO3]

a) Production being triggered by actual customer orders.
b) Manufacturing products based on forecasted future demand.
c) Prioritizing economies of scale over inventory control.
d) Maximizing efficiency for low demand uncertainty

34. Technological integration like ERP C EDI build trust by: [E]/[CO4]
a) Limiting information flow b) Improving data accuracy visibility
c) Increasing lead times d) Encouraging independent decisions

35. Consider the following statement and choose the most appropriate alternative: [E]/[CO3]
Statement I: Global supply chains are primarily designed to maximize efficiency (low cost) rather than responsiveness (speed/flexibility).
Statement II: A global supply chain that focuses on responsiveness typically utilizes inventory buffers and fast, expensive transportation modes.

a) Both statements are true
b) Statement I is true, and Statement II is false.
c) Statement I is false, and Statement II is true.
d) Both statements are false

36. The perfect order rate measures: [U]/[CO1]

a) Order delivered on time, complete C error free
b) The number of orders cancelled
c)Cost of returns
d) Number of suppliers

37. In the Barometric method, indicators that change before the economy or demand changes are classified as: [U]/[CO2]

a) Lagging indicators
b) Coincident indicators
c) Independent indicators
d) Leading indicators

38. For products with a short-expected delivery time (high customer expectation for Response Time), the necessary distribution network design would be a: [E]/[CO3]

a) Centralized network with few warehouses.
b) Decentralized network with many warehouses located close to customers.
c) Hybrid network focusing on low-value items.
d) Direct distribution model only.

39. Reverse logistics in GSCM refers to: [E]/[CO4]

a) Sending products only to retailors
b) Returning goods for recycling or reuse
c)Forward distributions only
d) Eliminating customer feedback

40. In inventory control process what is the term used for minimum level of stock that triggers new order? [E]/[CO2]

a) Safety stock
b) Cycle count
c)Economic Order Quantity
d) Reorder point

41. Consider the following statement of Assertion(A) and Reasoning (R). Choose the correct option. [E]/[CO2]

Assertion(A): Continuous flow supply chain is suitable for industries with stable demand and long product life.
Reasoning (R): The Supply Chain focus on maintaining efficiency and consistency in operations.

a) Both A and R are true, and R is the correct explanation of A
b) Both A and R are true, but R is not the correct explanation of A
c) A is true, but R is false
d) A is false, but R is true

42. Which of the following is not one of the key inputs that MRP process relies on? [E]/[CO2]

a) Master production schedule b) Inventory status record

c)Sales agents Composite d) Bill of materials

43. Match the following: [E]/[CO3]

(a) Weight and Volume	(i) Requires fast transit, often specialized (e.g., refrigerated trucks
(b) Perishability	(ii) Often justifies higher cost of faster, more secure transport
(c) Hazardous Materials	(iii) Heavier/bulkier products are cost-effective via sea or rail
(d) Value	(iv) Involves strict regulations and special carriers

Options:

A. a-i, b-iii, c-iv, d-ii

B. a-iii, b-i, c-iv, d-ii

C. a-iv, b-i, c-iii, d-ii

D. a-iii, b-i, c-ii, d-iv

44. The main role of Sourcing in SCM is to: [E]/[CO4]

a) Increase product price b) Ensure availability of quality materials

c)Delay production d) Reduce supplier relationships

45. Primary benefit of demand forecasting is that optimize stock levels, prevent stockouts and reduce carrying cost of excess inventory, which falls under: [E]/[CO2]

a) Production planning b) Inventory management

c)Financial Planning d) Strategic planning

46. Which of the following best describes a responsive supply chain? [U]/[CO1]

a) It focuses on low-cost and high efficiency b) Focus on flexibility and quick reactions to markets

c) Eliminates supplier collaborations. d) Ignores customer demand

47. Lead time is defined as: [U]/[CO2]

a) The time it takes to produce an item
b) The time between placing an order and receiving it
c) The time to sell an item
d) The time to inspect incoming goods

48. Consider the following statement of Assertion(A) and Reasoning(R). Choose the correct option. [E]/[CO3]

Assertion (A): Flexibility is the ability of a transport system unexpected event. to changes in demand

Reasoning (R): Road transport offers great flexibility for toute schedule changes compared to other modes.

a) Both A and R are true and R is the correct explanation of A
b) Both A and R are true but R is not the correct explanation of A
c) A is true but R is false
d) A is false but R is true

49. In the context of "Resilience," what does "multi-sourcing" mean? [E]/[CO4]

a) Relying on one supplier for cost efficiency
b) Sourcing from multiple suppliers to reduce dependency
c) Using only local suppliers
d) Buying from competitors